

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

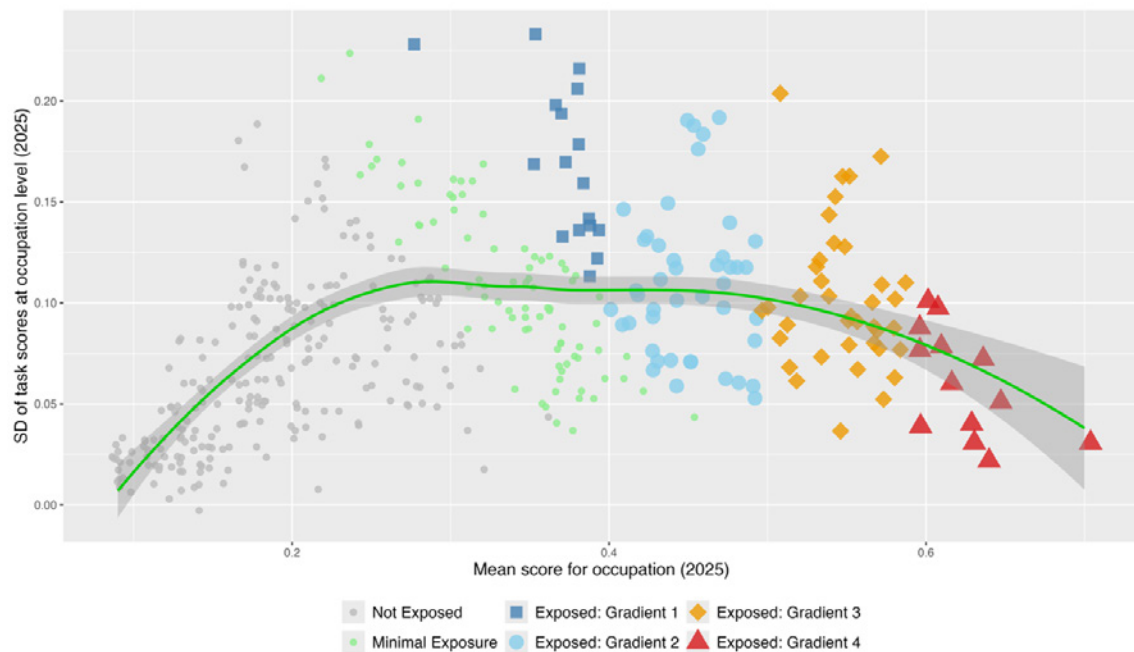
Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.

► 8 Adjusted Global Index of GenAI Exposure

To classify ISCO-08 occupations into varying levels of exposure to Generative AI (GenAI), we update the framework introduced in Gmyrek et al. (2023). Similarly to the original approach, we rely on two moments of distribution: occupational mean and standard deviation (SD) of task-level scores pertaining to a given 4-digit occupation. However, instead of using automation and augmentation potential as two extreme categories with the "big unknown" in between, we now adopt a more nuanced classification scheme that represents a spectrum of exposure, with occupations considered as exposed classified into four increasing gradients (Figure 16 and Table 5).

► Figure 16. AI Exposure Gradients: ISCO-08 4-digit level occupations



Note: Figure 16 presents all ISCO-08 4-digit occupations classified into GenAI exposure categories as per the criteria in Table 5. A detailed list of occupation-level means and SDs is available in the Appendix, Table A1. In addition, task-level scores within the overall hierarchy of ISCO-08 are available here: https://pgmyrek.github.io/2025_GenAI_scores_ISCO08/

Gradient 1 represents occupations with low overall GenAI exposure but significant variability across tasks (Low exposure, high task variability). While some tasks within these roles may have high automation potential, the occupation has many tasks that continue requiring human roles, aligning closely with the notion of augmentation in the old framework. Gradient 2 includes occupations with moderate GenAI exposure and a mix of highly exposed and minimally exposed tasks, resulting in uneven impacts where some tasks may be disrupted while others remain unaffected (Moderate exposure, high task variability). Gradient 3 captures occupations where a significant portion of tasks are consistently exposed to GenAI, signaling growing automation risks and requiring adaptation strategies for workers (Significant exposure, high task variability). Gradient 4 highlights occupations with the highest share of tasks exposed to potential GenAI-driven automation, and with a high consistency of this exposure across tasks (Highest exposure, low task variability). This gradient aligns closely with the notion of automation risk in our earlier classification system (Gmyrek et al., 2023). In addition, among occupations outside these four categories, we introduce a more granular distinction of Minimal Exposure, where some interaction with GenAI

► Conclusion

In this paper, we have presented the methodology used to develop a new global index of occupational exposure to GenAI, as well as new estimates of the global effects on employment. We build on the approach developed by the ILO in 2023, but with several notable refinements. First, by beginning our analysis with the Polish 6-digit occupational structure, we expand by ten-fold the amount of detailed task information available for the analysis of task automation. In addition, using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people in Poland, we captured the opinions of each 1-digit ISCO-08 group on the potential for task automation in the job categories in which they are currently employed (52,558 data points). We then compared this input with an expert survey and conducted subsequent rounds of revisions among a smaller international expert group for the tasks with the highest degree of disagreement. Based on this process, we built a repository of knowledge with adjusted task scores that goes beyond Poland's national specificities and reflects a broader consensus on the task automation potential of current GenAI tools.

We subsequently used this repository to develop an AI assistant that could predict scores for these tasks based on this additional information as well as the technical documentation of ISCO-08. These scores update the original set of scores produced by the ILO in 2023 (Gmyrek et al., 2023). The use of a verified “knowledge base” presents an important methodological improvement, which greatly enhances stability of our predictions across independent LLMs (see Section 6), and which can be used to produce similar score predictions for other occupational classification systems that contain job-specific tasks. In addition, we make all task-level scores publicly available, which also makes it possible for researchers to organize the occupational exposure in different ways than our proposed framework of exposure gradients.

The new global index presented in this paper is generic, due to the nature of ISCO-08 and the structure of tasks associated with occupations in this framework. Its definite strength – similar to ISCO-08 – is that it provides a common global denominator and a direct link to national labour forces surveys held in the ILO harmonized micro data repository, which relies on the same classification of occupational structures. Even though all countries do not use ISCO-08, mappings of the national systems to this common international standard are available. This eliminates the need for crosswalks from other national systems (e.g. O*NET), which involves tenuous assumptions about the similarity of detailed occupations in the technologically advanced context of the USA to other countries, especially non-HICs. The simple and comparable character of our ISCO-08 based index enables direct production of global, regional and income-based employment estimates, based on the same statistical modelling techniques that are used by the ILO for other global figures (see Section 11).

Our 2025 scores are presented in a revised occupational classification system, adopting a framework of four progressively increasing exposure gradients, that better highlights the importance of potential AI exposure and task variability for determining possible outcomes. We produced a new set of global estimates of GenAI exposure based on this revised classification.

The new index maintains many of the headline findings of the 2023 methodology. Clerical occupations continue to have the highest exposure levels, albeit with some important downward revisions to individual tasks scores. Such occupations include data entry clerks, typists, accounting, bookkeeping clerks and administrative secretaries. Additionally, some occupations – such as financial analysts, and web and multimedia developers (gradient 4), as well as bank tellers, translators, applications programmers, investment advisers (gradient 3) – have increased their